Energy System Modeling - Postdoctoral Researcher

Lawrence Livermore National Laboratory · Livermore, CA, United States

Job ID (this archive)	ENERGY-123
Employer	Lawrence Livermore National Laboratory
Location(s)	Livermore, CA, United States
Career stage (archive tag)	Graduate degree / research (Postdoctoral)
Location group	Other U.S.
Majors (archive tags)	Data Science, Computational & Geospatial Science
Energy area	Energy systems / computational modeling
Pay listed	$122,028 – $143,328 per year
Posted / listing date	46286.99696894676
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://jobs.smartrecruiters.com/LLNL/3743990015543786-energy-system-modeling-postdoctoral-researcher
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-094

> Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

Company Description

Join us and make YOUR mark on the World!

Lawrence Livermore National Laboratory (LLNL) has turned bold ideas into world-changing impact advancing science and technology to strengthen U.S. security and promote global stability.

Our mission spans four critical national security areas nuclear deterrence, threat preparedness, energy security, and multi-domain defense empowering teams to take on the toughest challenges of today and tomorrow. With a culture built on innovation and operational excellence, LLNL is a place where your expertise can make a real impact.

Job Description

We have an opening for a Postdoctoral Researcher to conduct basic and applied research in the continuum modeling, simulation, and optimization of energy and climate systems. You will work independently and as part of multidisciplinary teams to both apply existing commercial tools and develop custom physics-based modeling, simulation, and optimization codes in focus areas including energy generation, harvesting, storage, and conversion and carbon capture, storage, and conversion, and advanced manufacturing. This position is in the Computational Engineering Division (CED), within the Engineering Directorate.

Depending on your assignment, this position may offer a hybrid schedule, blending in-person and virtual presence. You may have the flexibility to work from home one or more days per week.

You will

- Conduct research and development in energy materials and devices.
- Develop continuum models to describe the fluid flow, mass and energy conservation, and chemical reactions describing energy systems (e.g., electrochemical reactors, batteries, fuel cells, capacitors, heat exchangers, absorbers, etc.).
- Employ and extend continuum commercial, open-source, and/or custom LLNL simulation codes to model electrochemical systems.

- Develop and implement High-Performance Computing (HPC) algorithms and simulations for the optimal design of engineering systems governed by nonlinear, transient, multiscale, and coupled physical phenomena.
- Work with engineers and scientists to identify and solve challenging problems providing contributions on modeling, simulation, and design optimization.
- Pursue independent (but complementary) research interests and interact with a broad spectrum of scientists internal and external to the Laboratory. Collaborate with others in a multidisciplinary team environment to accomplish research goals.
- Publish research results in peer-reviewed scientific or technical journals and present results at external conferences seminars and/or technical meetings.
- Perform other duties as assigned.

Qualifications

- PhD in engineering, mathematics, computational science, or a related field.
- Experience developing or modifying models and/or simulations using continuum (finite-element/finite-volume) approaches for a broad range of physical systems such as flowing reactive mass transfer systems (e.g., chemical reactors, electrochemical reactors, fuel cells, combustion, etc.), electrochemical phenomena, conjugate heat transfer, mass transfer, and structural mechanics.
- Experience programming in C/C++ and/or scripting languages like Python/Mathematica/MATLAB.
- Experience with commercial (e.g., COMSOL, STAR-CCM+, Ansys/Fluent, etc.), open-source (e.g., OpenFOAM, FEniCS, Firedrake, etc.), and/or other common modeling, simulation, and optimization codes.
- Ability to develop independent research projects as demonstrated through publication of peer-reviewed literature.
- Proficient verbal and written communication skills needed to effectively collaborate in a multidisciplinary team environment, present and explain technical information, document work, and prepare and present successful proposals and high-quality research papers.

Qualifications We Desire

- Detailed knowledge of continuum-scale models for simulating electrochemical systems.
- Experience with developing methods for structural and fracture mechanics coupled to electrochemical phenomena.
- Experience with homogenized and/or resolved formulations of partial differential equations (PDEs) for flow and transport in porous materials.
- Experience with explicit design sensitivity analysis via direct (forward) and adjoint (backward) methods applied to advanced shape/topology/design optimization techniques and algorithms.

Pay Range

$122,028 - $143,328 Annually

This is the lowest to highest salary we in good faith believe we would pay for this role at the time of this posting; pay will not be below any applicable local minimum wage. An employee’s position within the salary range will be based on several factors including, but not limited to, specific competencies, relevant education, qualifications, certifications, experience, skills, seniority, geographic location, performance, and business or organizational needs

Additional Information

#LI-Hybrid

Position Information

This is a Postdoctoral appointment with the possibility of extension to a maximum of three years, open to those who have been awarded a PhD at time of hire date.

Why Lawrence Livermore National Laboratory?

- Included in 2026 Best Places to Work by Glassdoor!
- Flexible Benefits Package
- 401(k)
- Relocation Assistance
- Education Reimbursement Program
- Flexible schedules (*depending on project needs)
- Our values - visit https://www.llnl.gov/inclusion/our-values

Security Clearance

This position requires either no security clearance, or a Department of Energy (DOE) L-level or Q-level clearance depending on the particular assignment.

If you are selected and a security clearance is required, we will initiate a Federal background investigation to determine if you meet eligibility requirements for access to classified information or matter. Also, all L or Q cleared employees are subject to random drug testing. L and Q-level clearances require U.S. citizenship.

If no security clearance is required, but your assignment is longer than 179 days cumulatively within a calendar year, you must go through the Personal Identity Verification process. This process includes completing an online background investigation form and receiving approval of the background check.

National Defense Authorization Act (NDAA)

The 2025 National Defense Authorization Act (NDAA), Section 3112, generally prohibits citizens of China, Russia, Iran and North Korea without dual US citizenship or legal permanent residence from accessing specific non-public areas of national security or nuclear weapons facilities. The restrictions of NDAA Section 3112 apply to this position. To be qualified for this position, Candidates must be eligible to access the Laboratory in compliance with Section 3112.

Pre-Employment Drug Test

External applicant(s) selected for this position must pass a post-offer, pre-employment drug test. This includes testing for use of marijuana as Federal Law applies to us as a Federal Contractor.

Wireless and Medical Devices

Per the Department of Energy (DOE), Lawrence Livermore National Laboratory must meet certain restrictions with the use and/or possession of mobile devices in Limited Areas. Depending on your job duties, you may be required to work in a Limited Area where you are not permitted to have a personal and/or laboratory mobile device in your possession. This includes, but not limited to cell phones, tablets, fitness devices, wireless headphones, and other Bluetooth/wireless enabled devices.

If you use a medical device, which pairs with a mobile device, you must still follow the rules concerning the mobile device in individual sections within Limited Areas. Sensitive Compartmented Information Facilities require separate approval. Hearing aids without wireless capabilities or wireless that has been disabled are allowed in Limited Areas, Secure Space and Transit/Buffer Space within buildings.

How to identify fake job advertisements

Please be aware of recruitment scams where people or entities are misusing the name of Lawrence Livermore National Laboratory (LLNL) to post fake job advertisements. LLNL never extends an offer without a personal interview and will never charge a fee for joining our company. All current job openings are displayed on the Career Page under “Find Your Job” of our website. If you have encountered a job posting or have been approached with a job offer that you suspect may be fraudulent, we strongly recommend you do not respond.

To learn more about recruitment scams:
https://www.llnl.gov/sites/www/files/2023-05/LLNL-Job-Fraud-Statement-Updated-4.26.23.pdf

Equal Employment Opportunity

We are an equal opportunity employer that is committed to providing all with a work environment free of discrimination and harassment. All qualified applicants will receive consideration for employment without regard to race, color, religion, marital status, national origin, ancestry, sex, sexual orientation, gender identity, disability, medical condition, pregnancy, protected veteran status, age, citizenship, or any other characteristic protected by applicable laws.

Reasonable Accommodation

Our goal is to create an accessible and inclusive experience for all candidates applying and interviewing at the Laboratory. If you need a reasonable accommodation during the application or the recruiting process, please use our online form to submit a request.

California Privacy Notice

The California Consumer Privacy Act (CCPA) grants privacy rights to all California residents. The law also entitles job applicants, employees, and non-employee workers to be notified of what personal information LLNL collects and for what purpose. The Employee Privacy Notice can be accessed here.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official SmartRecruiters job data reports active=true and supplies a job-specific apply URL. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: PhD in engineering, mathematics, computational science, or a related field. | This is a Postdoctoral appointment with the possibility of extension to a maximum of three years, open to those who have been awarded a PhD at time of hire date.

Key skills listed in the archive: Python; C++; MATLAB; optimization

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer SmartRecruiters Employer / ATS identifier: REF8882R

Career stage: Postdoctoral

Major fit: Advanced computational pathway: energy-system modeling; relevant PhD required. Energy-sector connection: Modeling of energy systems and decarbonization.

Retrieval note: New posting found in expansion round. Official SmartRecruiters job data reports active=true and supplies a job-specific apply URL. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer SmartRecruiters. Source identifier: REF8882R.

Sources and retrieval details

Employer posting

https://jobs.smartrecruiters.com/LLNL/3743990015543786-energy-system-modeling-postdoctoral-researcher

Retrieved source

https://api.smartrecruiters.com/v1/companies/LLNL/postings/3743990015543786

Retrieved: 2026-10-26T16:35:32.639910+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.